Master's Seminar examination brief

Spatiotemporal Modeling of Citizen Tree Watering in Berlin

Weather Responses and Neighborhood Variation

Joseph Carpenter Sheils
Saarland University | M.Sc. Data Science and Artificial Intelligence
Supervisor: Angela John | Prepared 17 October 2026

Research purpose

The thesis examines how logged citizen care for Berlin's trees responds to heat and dryness, and how that response varies across neighbourhoods. The existing data and exploratory results demonstrate a workable foundation. The core daily response study remains to be fitted.

Study	Role and planned contribution
Part II: civic heat response	Core. Daily logged watering, weather response, social context and the rival explanation of participation inequality.
Part I: vegetation cooling loss	Supporting. Morning surface-temperature relationships with canopy, low vegetation and temporal drought.
Part III: allocation	Exploratory. Validated prediction of logged care and comparison of need-only versus complement-based priorities.

What is already available

57,786 linked records | 11,127 distinct trees | 2,249,502 recorded litres
542 planning areas | 68 observed months | 50,259 positive tree-days

The package includes the October proposal, seminar slides, detailed results, twelve map plates, source snapshots, derived tables and spatial layers, code, and verification instructions. The tree inventory is supplied as a lossless compressed snapshot.

Scope: preliminary and descriptive. Logged activity is not total irrigation, response is not causal cooling, and screening candidates are not allocation recommendations.

Evidence and sample reconciliation

The canonical analysis window is May 2020 through December 2025. May 2020 is partial. The complete-year temporal atlas uses 2021-2025.

Sample stage	Records / interpretation
Raw snapshot through April 2026	58,847 records; 11,278 source tree IDs
After exact duplicates	58,736 records; 111 duplicate rows removed
Study window	58,474 events; 2,277,987 litres before linkage
Inventory linked and LOR assigned	57,786 events; 11,127 trees; 2,249,502 litres
Unmatched or unassigned	688 events; 28,485 litres excluded from geographic panel

Preliminary findings

Seasonality. 71.6% of linked events occur in June-August and 92.2% in May-September. This motivates explicit seasonal adjustment in the proposed daily study.

Spatial structure. Global Moran's I for reach is 0.441, with permutation $p = 0.001$. Local classifications use unadjusted $p < .05$ and are exploratory.

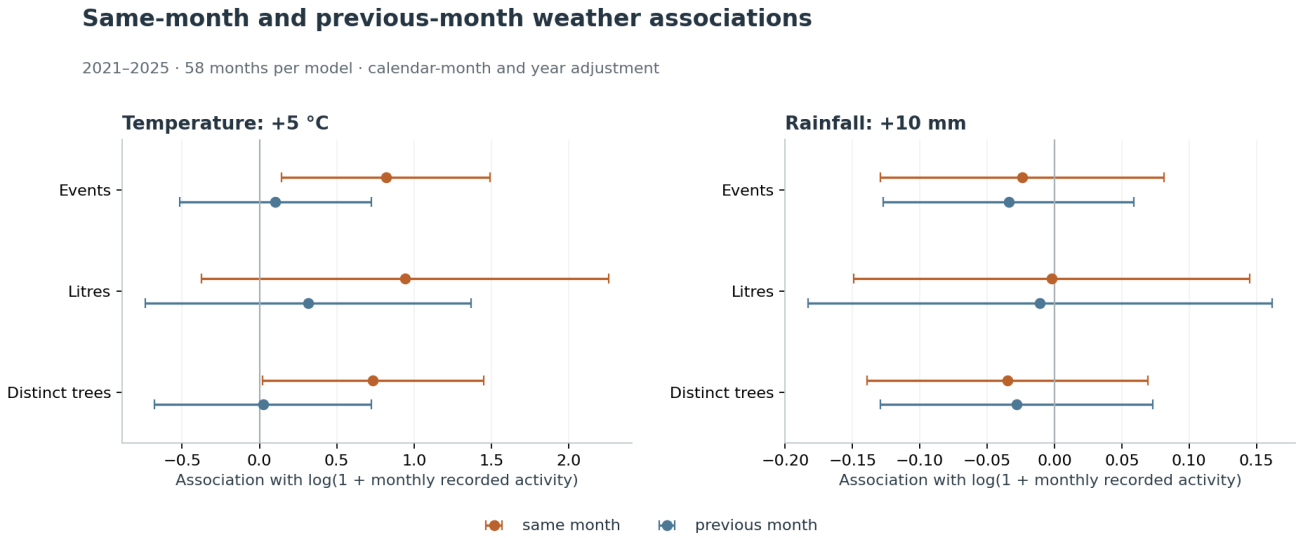
Different dimensions of need. Across 535 valid observations, reach correlates weakly negatively with child poverty (rho -0.123) and positively with the environmental/social burden count (rho 0.486). Middle-status areas have the highest mean monthly event rate; there is no simple monotonic advantage gradient.

Count qualification: 11,160 is a full-snapshot inventory match count, not the canonical study-window LOR count. The older 11,087 and 11,245 figures in the proposal are not adopted. Invalid source coordinates alone do not define exclusion from the ID-linked panel.

Evidence: examination/preliminary_results.md links to foundation and tree-day audits, global Moran statistics, equity associations, and the complete model grid. All files are included in the repository.

Weather response and source limits

The same-month event association is 0.817 log(1 + events) units per +5 degrees C, with approximate 95% interval [0.142, 1.492]. The previous-month estimate is 0.103 [-0.515, 0.722]. These are exploratory monthly associations, not daily response slopes or causal effects.



The model jointly includes temperature and rainfall with calendar-month and year adjustment. It uses 58 of 60 months in 2021-2025; June and October 2023 lack complete rainfall. HAC(3) uncertainty preserves calendar gaps. Intervals are not multiplicity-adjusted.

Two temperature measurements

DWD Berlin-Tempelhof provides shared station air-temperature context. The supplied monthly Ortsteil file describes surface temperature, with different support and unresolved processing. These measurements are not interchangeable.

The Ortsteil extension has 6,596 area-months and 93 provisionally comparable areas. Of 9,312 source rows, 1,811 have missing LST. Buckow components and changed boundaries require resolution. Scene dates, cloud filtering, compositing, rainfall provenance and count semantics were not supplied.

The next-month litres/LST residual correlation changes sign under sensitivity checks (-0.024 overall seasonal sample, +0.005 without 2022, +0.075 over all months). It provides no stable descriptive cooling pattern and does not validate a causal mechanism.

Sources: output/watering_response/report.html; tables/monthly_adjusted_associations.csv; ortsteil/tables/provisional_pooled_correlations.csv. The report supplies all exploratory specifications, not only selected associations.

Feasibility and next decisions

The October proposal sets a six-month programme, relative to an agreed thesis start. Prioritise the core behavioural study, then the supporting physical analysis, then the exploratory decision layer.

Period	Planned output or decision
Month 1	Literature and data audit; source requests; offsets and participation proxies; prospective registration with prior exploration disclosed
Months 1-2	Balanced behavioural area-day panel and satellite/vegetation panel
Months 2-4	Core response model, participation explanations and temporal stability
Months 3-4	Cooling-loss model; end-Month-4 decision on full versus descriptive allocation analysis
Month 5	Validated prediction and allocation comparison if feasible
Month 6	Bridge test where both layers exist, scale checks, writing and code release

Issues to resolve before substantive inference

Build historical young-tree denominators and observed zero days; complete daily weather, calendar and participation controls; clarify the supplied LST extraction; disclose already-inspected outcomes in any registration; validate prediction and calibration before ranking areas.

Source access and reproducibility

The package provides hashes, CSV schemas, source attribution, a tree-inventory restore command, and tests. The supplied monthly LST and legacy borough inputs have unresolved provenance or licence metadata. Proposed satellite, vegetation-height, population-age, turnout and other control layers are not presented as already available.

Read examination/README.md for the entry point, examination/reproducibility.md for ordered commands, and examination/validation.md for the actual verification performed. The supplied proposal is preserved unchanged and has a separate qualification note.

This evidence supports an early feasibility review. It does not certify institutional approval or completion of the planned models.